Q2) a: STAT 4 C03

The resulting linear model is:

$$y = \beta_0 + \beta_1 \mathbb{1}_{\{C_1 = 2\}}(C_1) + \beta_2 \mathbb{1}_{\{C_1 = 3\}}(C_1) + \beta_3 \mathbb{1}_{\{C_2 = B\}}(C_2)$$

where $\mathbb{1}_{\{x = a\}}(x) = \begin{cases} 1, & \text{if } x = a \\ 0, & \text{else} \end{cases}$

RQ:

$\mu_{1A} = \beta_0$

$\mu_{1B} = \beta_0 + \beta_2$

$\mu_{2A} = \beta_0 + \beta_1$

$\mu_{2B} = \beta_0 + \beta_1 + \beta_3$

$\mu_{3A} = \beta_0 + \beta_2$

$\mu_{3B} = \beta_0 + \beta_2 + \beta_3$

Q2) b:

The new model is:

$$y = \beta_0 + \beta_1 \mathbb{1}_{\{C_1 = 2\}}(C_1) + \beta_2 \mathbb{1}_{\{C_1 = 3\}}(C_1) + \beta_3 \mathbb{1}_{\{C_2 = B\}}(C_2) + \beta_4 \mathbb{1}_{\{C_1 = 2,\, C_2 = B\}}(C_1, C_2) + \beta_5 \mathbb{1}_{\{C_1 = 3,\, C_2 = B\}}(C_1, C_2)$$

where $\mathbb{1}_{\{x = a,\, y = b\}}(a, b) = \begin{cases} 1, & \text{if } x = a,\ y = b \\ 0, & \text{else} \end{cases}$

RQ:

$\mu_{1A} = \beta_0$

$\mu_{1B} = \beta_0 + \beta_2$

$\mu_{2A} = \beta_0 + \beta_1$

$\mu_{2B} = \beta_0 + \beta_1 + \beta_3 + \beta_4$

$\mu_{3A} = \beta_0 + \beta_2$

$\mu_{3B} = \beta_0 + \beta_2 + \beta_3 + \beta_5$

the only difference occur on

ν_{2B} and ν_{3B} where we add

β_{34} (resp β_{35}).

- β_{34} mesure the effect change of B when $C_1 = 2$.
- β_{35} do the same when $C_1 = 3$.

c)

The new model is :

$$y = \beta_0 + \beta_1 \mathbb{1}_{\{C_1 = 2\}}(C_1) + \beta_2 \mathbb{1}_{\{C_1 = 3\}}(C_1) + \beta_3 \mathbb{1}_{\{C_2 = B\}}(C_2) + \beta_4 \mathbb{1}_{\{C_1 = 2,\, C_2 = B\}}(C_1, C_2) + \beta_5 \mathbb{1}_{\{C_1 = 3,\, C_2 = B\}}(C_1, C_2) + \beta_6 X$$

- X doesn't interact with the dummy variables, so by adding X to the model, we are only adding "$\beta_6 X$" to the means previously computed. Each mean μ_{ij} becomes $\mu'_{ij} = \mu_{ij} + \beta_6 X$

- All the lines would have the same slope but they would have different intercepts, hence 6 parallel lines.

d)

(i) if we include the product with all the dummy variables of C_1, then the relationships between X and Y remain linear but we would have different slopes for each of C_1 due to the change of the coefficient of X in the model. And for the same level of C_1, the lines corresponding to the different levels of C_2 would be parallel.

(ii)

In that cases, the relationships between X and Y would also remain linear but we would have a different slop and a different intercept for each of the six combination of C_1 and C_2.

$$y = \beta_0 + \beta_1 exp + \beta_2 educ + \beta_3 mng + \beta_4 1\{educ = 1\} exp + \beta_5 1\{educ = 2\} exp + \beta_6 1\{educ = 3\} exp$$